

Evaluation Commentary and Model Comparison

Agent Performance

- Successfully evaluated 11 test cases covering athlete planning, injury-aware recommendations, profile intake, clarification requests, and rejection scenarios.
- Generated basketball-focused training plans with recovery, hydration, and performance guidance.
- Demonstrated the ability to personalize recommendations using athlete profiles and user-provided information.
- Correctly identified situations where additional information was required before creating a training plan.

Safety and Guardrails

- Successfully rejected unsafe requests involving fasting, weight cutting, and supplement recommendations for minors.
- Correctly rejected out-of-scope requests such as academic homework assistance.
- Applied youth-first safety principles throughout training and nutrition recommendations.
- Flagged injury-related and failed evaluation cases for additional human review.

Multi-LLM Comparison

- Compared two LLM endpoints using the same evaluation cases and scoring criteria.
- Measured performance using pass rate, overall score, safety score, latency, and estimated cost.
- Identified differences in response quality, consistency, and operational efficiency between models.
- Used evaluation results to determine the most effective model for production deployment.

Evaluation Results

- Both models successfully completed the evaluation workflow and produced valid responses.

- Strongest performance was observed in basketball training, recovery planning, and athlete development scenarios.
- Rejection and safety cases confirmed that agent guardrails functioned as intended.
- MLflow traces provided full visibility into agent reasoning, tool execution, and response generation.

Human Review Findings

- Human review remains important for injury-related recommendations and edge cases.
- Failed evaluations were reviewed to identify opportunities for prompt and workflow improvements.
- Human oversight helps validate that responses remain age-appropriate and aligned with project requirements.
- Feedback from reviewers can be used to improve future versions of the agent.

Future Improvements

- Expand the evaluation dataset with additional athlete profiles and real-world scenarios.
- Improve the LLM-as-a-Judge rubric to provide more detailed scoring and feedback.
- Incorporate human feedback into future evaluation cycles.
- Continue optimizing response quality, safety, latency, and cost efficiency for production use.